DC-7 Drip Controller Field Trial, Q3 2026

Summary

Northwind Irrigation Cooperative ran a twelve-week field trial of the DC-7 drip controller from 6 July to 27 September 2026, on four member sites with twelve controllers each (48 units in all).

Across the four sites the trial plots used 15,380 cubic metres of water against a 2025 baseline of 20,000, a reduction of 23.1%. Every site improved; the smallest gain was at Cedar Ridge.

Two problems were found: valve stiction at Cedar Ridge and clock drift in firmware 2.3.1. Both have a fix, described in Section 4.

2. Method

The baseline is the same plots in the 2025 season, irrigated on fixed timers. Flow meters were read weekly by site stewards; soil moisture was logged at 30 cm by one capacitive probe per plot.

Weeks are numbered 1 to 12 from the week of 6 July. Weekly readings, soil moisture and controller alerts are in the companion spreadsheet, sensor-log.xlsx, sheet weekly_readings.

Rainfall during the trial was within 8% of the 2025 season at every site, so no weather correction was applied.

3. Results by site

Table 1. Water use by site, cubic metres.

Site	Units	Baseline 2025	Trial 2026	Reduction %
Alder Flat	12	4,820	3,610	25.1
Birch Hollow	12	5,140	4,010	22.0
Cedar Ridge	12	4,460	3,700	17.0
Dunmore	12	5,580	4,060	27.2
Total	**48**	**20,000**	**15,380**	**23.1**

Dunmore gained most (27.2%). Cedar Ridge gained least, and its result is depressed by the valve fault in Section 4.

4. Problems found

Valve stiction. At Cedar Ridge, 5 of the 12 controllers reported a valve that failed to close fully, mostly in week 7. The cause was scale from hard well water. An inline filter cures it; a retrofit is planned for every unit before any expansion.

Clock drift. Firmware 2.3.1 lost up to 9 minutes a week, which shifted watering windows into the afternoon. Firmware 2.3.4, installed on 18 August, fixed it; no drift was seen after that date.

5. Recommendation

Expand to 200 controllers for the 2027 season, conditional on the filter retrofit. The operations board decides at its meeting on 12 November 2026.

Contact for this report: the cooperative’s operations team.